

Section 1: Executive Summary

Write 2-3 paragraphs explaining:

- *What system you analyzed*
- *Why threat modeling matters for this system*
- *Your top 3 findings (highest priority threats)*

Write this for Marcus Webb, the General Manager. He's not technical. Use plain language.

- I analyzed the HYDROLOGIC water flow management system utilized at The Grand Marina, a 500 room luxury hotel. The system consists of three connected HYDROlogic devices and is installed to access critical water service areas. These devices monitor water flow, report data, and allow staff to remotely shut off water through a dashboard. Threat modeling is crucial for this system because it controls the physical infrastructure. An issue with the system won't just impact the technology but cause real world damage. The 4.2 million damage incident triggered because a leak alert was ignored showing that having secure, available, and reliable systems are business crucial. A cybersecurity attack on these systems may occur in similar or even worse circumstances. Thus, creating financial risk, operational shutdowns, regulatory exposure, and even reputational harm.
1. Unauthorized access causing system manipulation of water control
 2. Alert suppression or tampering
 3. Dashboard compromise

Section 2: System Overview

Describe The Grand Marina's HYDROLOGIC system:

- *500-room luxury hotel (Hydroficient customer)*
- *3 HYDROLOGIC flow management devices (one per water service line)*
- *Cloud-based monitoring and control*
- *Web dashboard for operators and management*
- *Remote shutoff and gate control capability*

Include a simple data flow diagram (can be ASCII art or describe the flow):

None

[HYDROLOGIC Devices] → [Cloud API] → [Dashboard]



[Gate Controls]



[Database]



Remote Gate

[Emergency Shutoff]

[Reports]

Section 3: Asset Inventory

List the key assets and their CIA priorities (use your work from Step 2):

Asset	Description	C Priority	I Priority	A Priority
HYDROLOGIC Devices	3 flow management units	Medium	High	High
Web Dashboard	Operator monitoring interface	High	High	Medium
Cloud API	Device-to-cloud communication	Medium	High	High
Remote Controls	Gate adjustments, emergency shutoff	Low	Critical	Critical
Consumption Data	Savings reports, billing records	Medium	Medium	Low

Section 4: STRIDE Analysis

This is the core of your threat model. For **each major component**, analyze all six STRIDE categories:

Component 1: HYDROLOGIC Devices

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	- Threat actor impersonates a device through stolen credentials	- Medium	- High	- High
Tampering	- Firmware modified to suppress leak alerts	- low	- Critical	- High
Repudiation	- No logging so there is no confirmation of anything	- Medium	- Medium	- medium
Info Disclosure	- Unencrypted traffic	- Medium	- Medium	- Medium
Denial of Service	- Flooding device with packets	- medium	- high	- high
Elevation of Privilege	- Firmware vulnerable	- low	- Critical	- high

	ility to get admin access			
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Component 2: Web Dashboard (Same table structure)

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	- Stolen credentials used to access dashboard	- Medium	- High	- High
Tampering	- Threat actor alters thresholds	- Medium	- high	- High
Repudiation	- No audit logs	- Medium	- Medium	- medium
Info Disclosure	- Leaked reports about hotel ops	- Medium	- Medium	- Medium
Denial of Service	- Dashboard unavailable during emergency	- medium	- high	- high
Elevation of Privilege	- Role misconfiguration	- low	- Critical	- Critical

Component 3: Cloud API (Same table structure)

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	- Leaked API keys and reused	- Medium	- High	- High
Tampering	- Tampered API keys and sends false data	- Medium	- high	- High
Repudiation	- Lack of logging of control actions	- Medium	- Medium	- medium
Info Disclosure	- Misconfiguration of storage allowing for data exposure	- Medium	- Medium	- Medium
Denial of Service	- API outage disables monitoring	- medium	- high	- high
Elevation of Privilege	- API flaw causing lateral movement	- low	- Critical	- High

Component 4: Remote Controls (Gate/Shutoff) (Same table structure)

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	- Unauthorized user issues shutoff	- Medium	- Critical	- Critical
Tampering	- Threat actor alters commands in transit	- low	- Critical	- High
Repudiation	- No traceability for shutoff events	- Medium	- Medium	- medium
Info Disclosure	- Leaked reports about hotel ops	- Low	- Low	- Low
Denial of Service	- Controls not available during active leak	- medium	- Critical	- Critical
Elevation of Privilege	- User bypasses approval	- Medium	- Critical	- Critical

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Section 5: Risk Summary

List your findings by risk level:

Critical Risks:

- 1. Unauthorized access to remote shutoff controls can cause flooding or outages
- 2. Dashboard account compromise can allow threat actor full access to full system control

High Risks:

- 1. Alert Suppression or Tampering

Medium Risks:

- 1. .Exposure of consumption data

Section 6: Recommended Mitigations

For each Critical and high-risk, propose a defense:

Risk	Proposed Mitigation	Implementation Complexity
Dashboard Compromise	Enable multi-factor authentication and RBAC	Low
Unauthorized Shutoff	- Multiple people approval for remote shutoff	- low
Alert tampering	Cryptographic signing of device alerts	- medium

